

Lab 10 - Report

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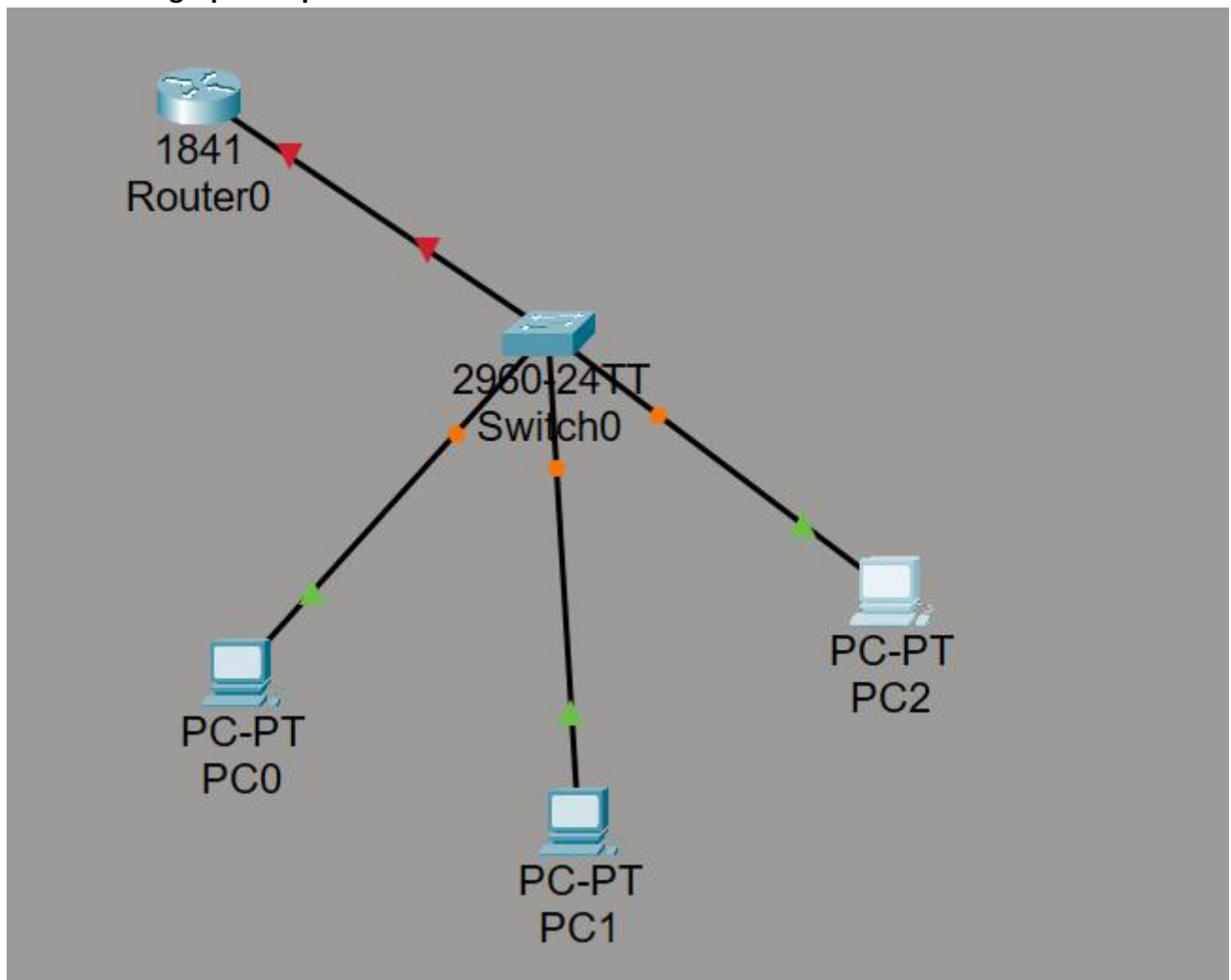
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Group: Mon 11:00-13:00

Practice 10 - Understanding and Configuring the DHCP Protocol

Part 1: Setting Up a Simple DHCP Network

Part 1: Setting Up a Simple DHCP Network



1.1 Router Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
ip
% Incomplete command.
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
```

1.2 Configuring DHCP Pool on Router

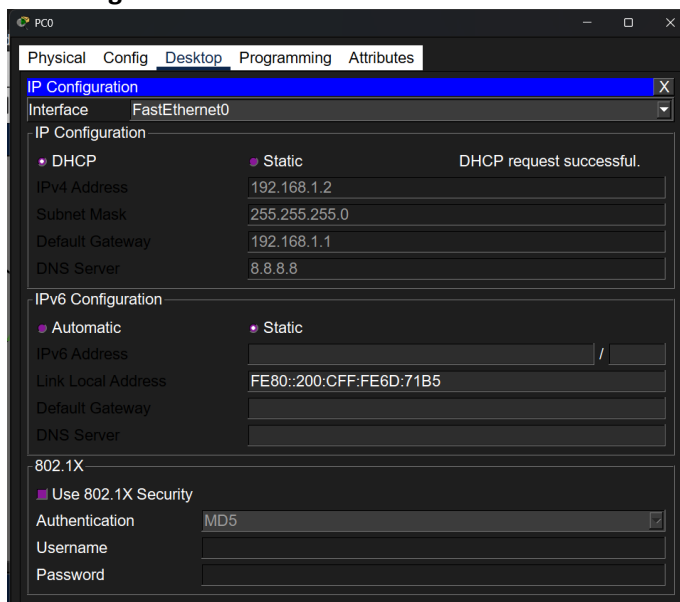
```
Router(config)#
Router(config)#ip dhcp pool DHCP
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
```

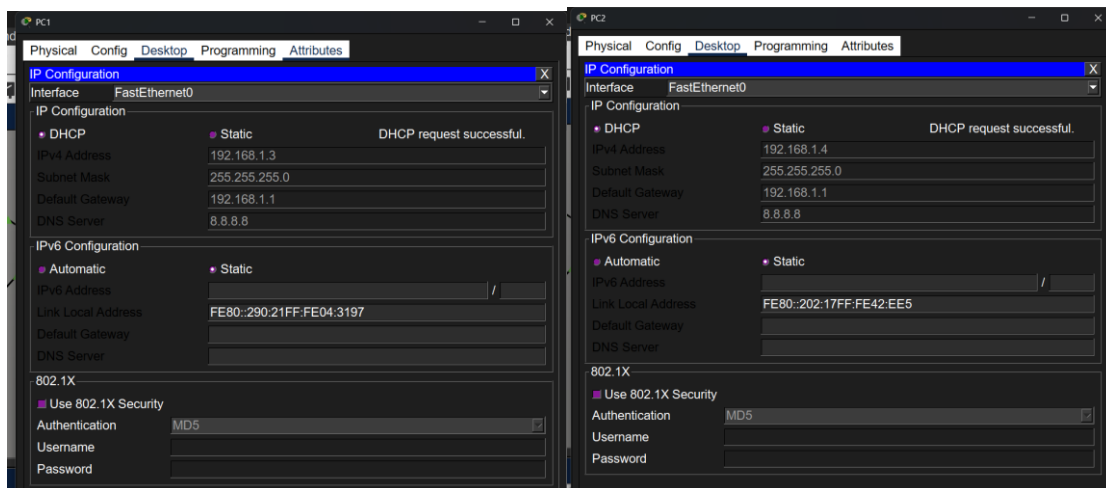
1.3 Excluding IP Addresses

```
Router(config)#
Router(config)#ip dhcp excluded-address 192.168.1.100
Router(config)#ip dhcp excluded-address 192.168.1.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

1.4 PC Configuration

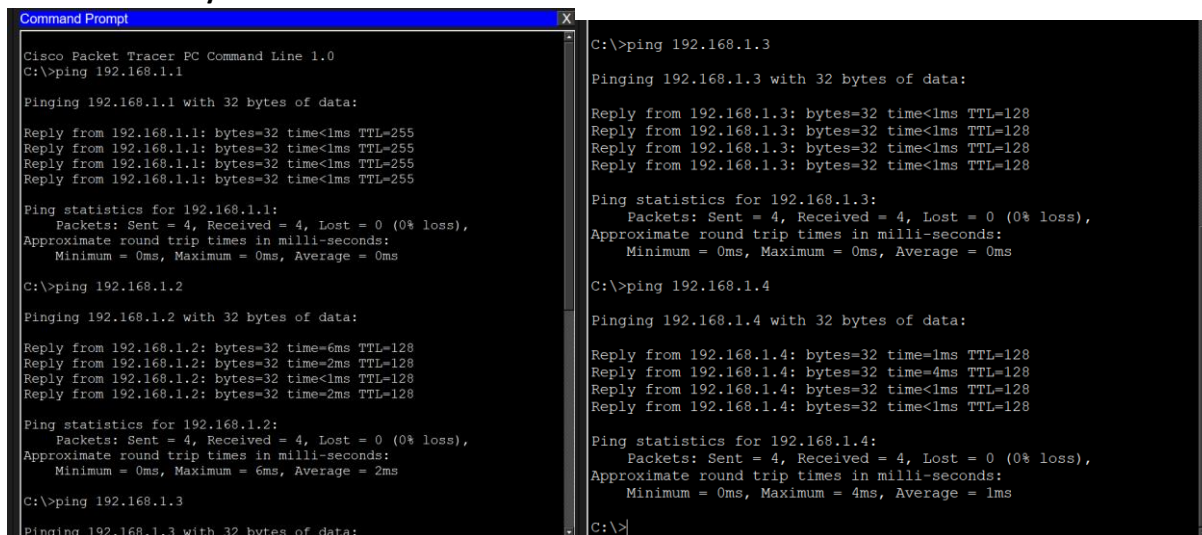




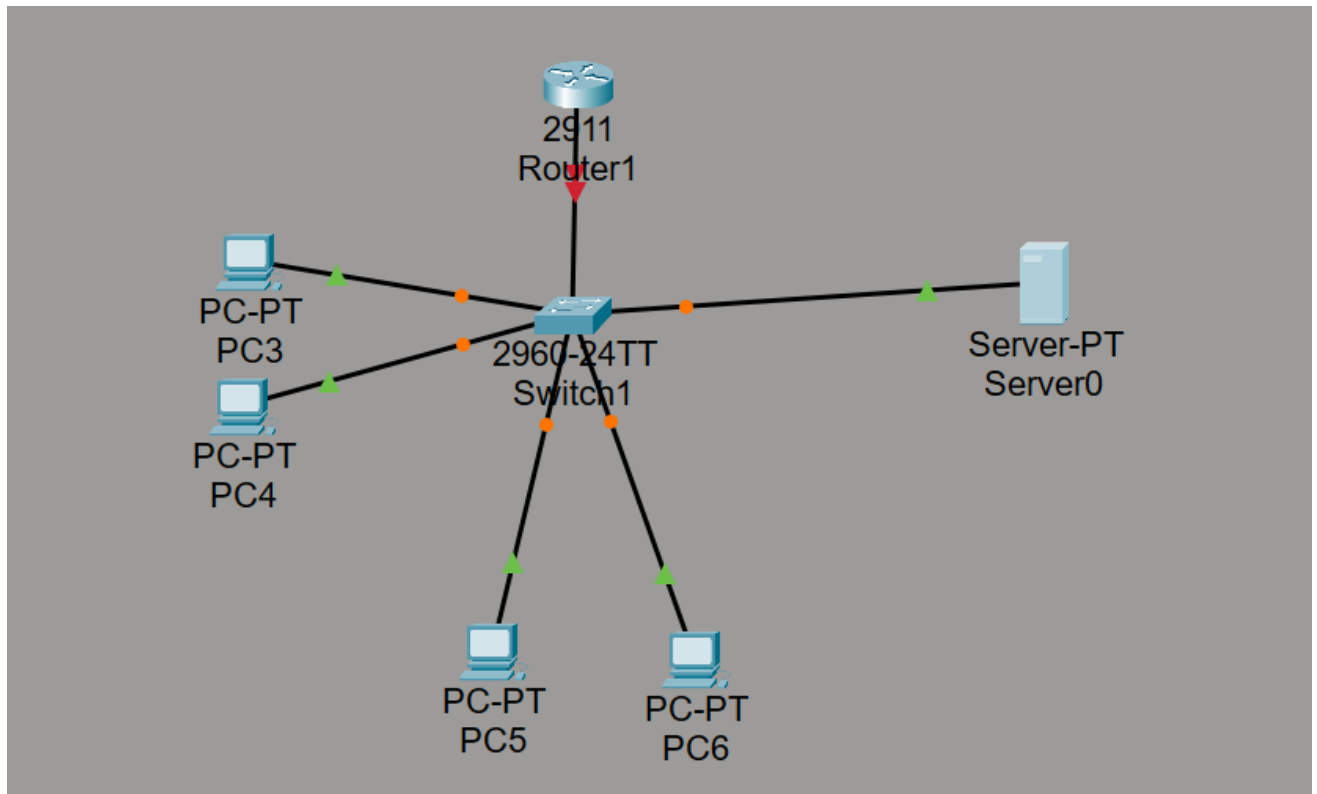
Display the list of IP addresses that have been dynamically assigned by the DHCP server:

```
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration
Type            Hardware address
192.168.1.2     0000.0C6D.71B5  --
Automatic
192.168.1.3     0090.2104.3197  --
Automatic
192.168.1.4     0002.1742.0EE5  --
Automatic
Router#
```

1.5 Test Connectivity



Part 2: VLAN Configuration with DHCP Relay



2.1 VLAN Configuration on the Switch

Create VLANs and Assign Names

```
Switch(config)#
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name DHCP
Switch(config-vlan)#exit
Switch(config)#
```

Assign VLANs to Switch Ports

```
Switch(config)#int range fastEthernet 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#int range fastEthernet 0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#exit
Switch(config)#int range fastEthernet 0/6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 4
Switch(config-if-range)#exit
Switch(config)#int fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3,4
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

2.2 Configuring Sub-Interfaces on the Router for InterVLAN Routing

2.2.1 Configure Sub-Interface for VLAN 2

```
Router(config)#int gigabitEthernet 0/0.2
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
Router(config)#int gigabitEthernet 0/0.3
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

2.2.3 Configure Sub-Interface for VLAN 4(DHCP VLAN)

```
Router(config)#
Router(config)#int gigabitEthernet 0/0.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

2.2.4 Enable Physical Interface GigabitEthernet 0/0

```
Router(config)#int gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.2, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.2, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.3, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.3, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.4, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.4, changed state to up

Router(config-if)#
```

2.2.5 Save Configurations

```
Router#wr mem
Building configuration...
[OK]
Router#
```

2.3 Configure DHCP Server

2.3.1 Assign IP Address to the Server

IP Configuration X

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

192.168.4.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.4.1

DNS Server

0.0.0.0

IPv6 Configuration

2.3.2 Configure DHCP for VLAN2

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

FastEthernet0

☒ ☐

DHCP-VLAN2

192.168.2.1

8.8.8.8

19216820

2552552550

256

0.0.0.0

0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192....	255....	512	0.0.0.0	0.0.0.0

Top

2.3.2 Configure DHCP for VLAN3

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

FastEthernet0

DHCP-VLAN3

192.168.3.1

8.8.8.8

192 168 3 0

255 255 255 0

256

0.0.0.0

0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192....	255....	512	0.0.0.0	0.0.0.0

Top

2.4 DHCP Relay Configuration

```
Router(config)#int gigabitEthernet 0/0.2
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#int gigabitEthernet 0/0.3
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#
```


2.5 Configuration IP Addresses on PCs.

The image displays four screenshots of PC configuration windows, labeled PC3, PC4, PC5, and PC6. Each window shows the 'Desktop' tab with the 'IP Configuration' section expanded. The configuration is for the 'FastEthernet0' interface.

PC3 Configuration:

- Interface: FastEthernet0
- IP Configuration: Static
- IPv4 Address: 192.168.2.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1
- DNS Server: 8.8.8.8

PC4 Configuration:

- Interface: FastEthernet0
- IP Configuration: DHCP
- Status: DHCP request successful.
- IPv4 Address: 192.168.2.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1
- DNS Server: 8.8.8.8

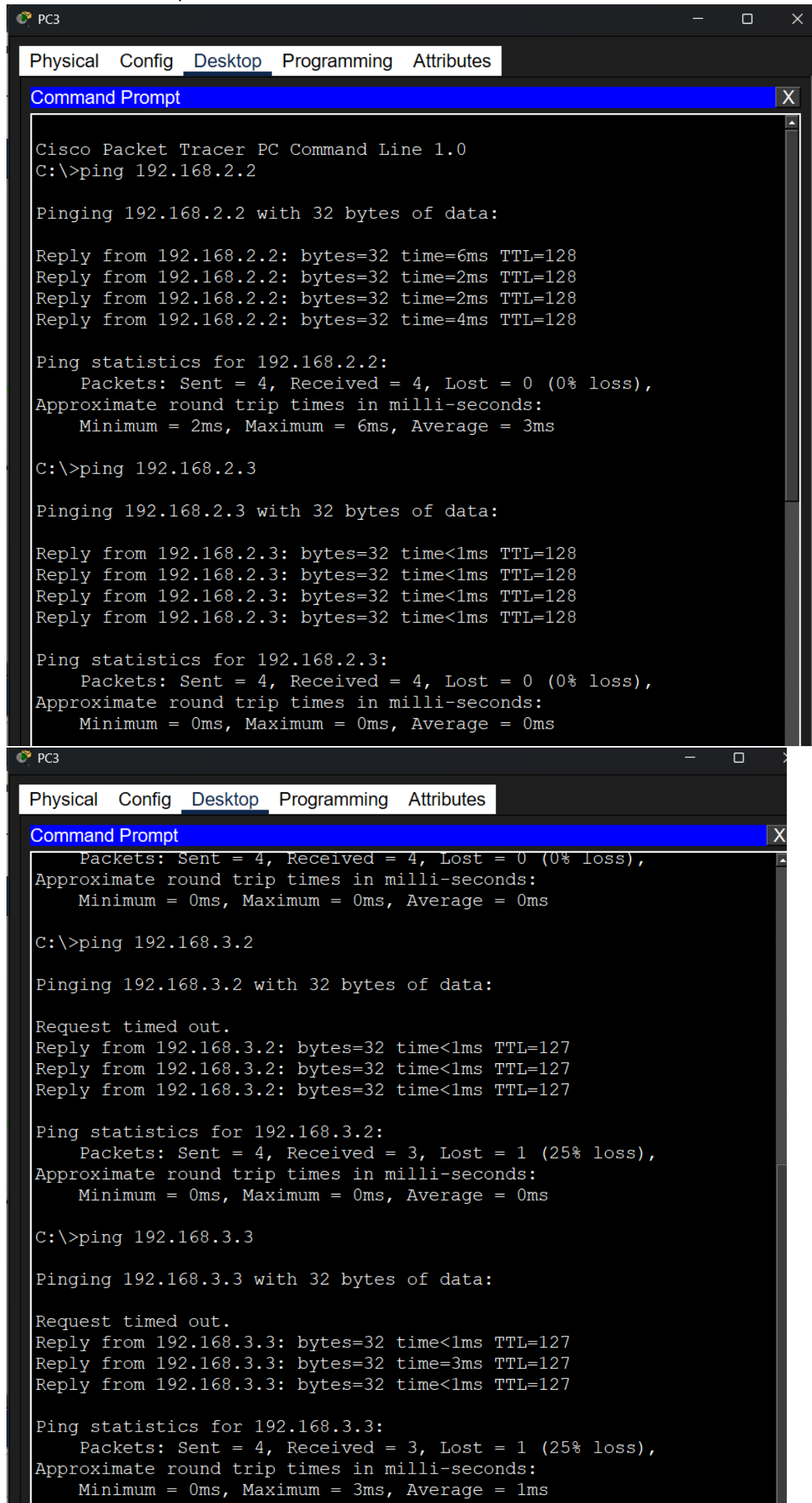
PC5 Configuration:

- Interface: FastEthernet0
- IP Configuration: DHCP
- Status: DHCP request successful.
- IPv4 Address: 192.168.3.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1
- DNS Server: 8.8.8.8

PC6 Configuration:

- Interface: FastEthernet0
- IP Configuration: DHCP
- Status: DHCP request successful.
- IPv4 Address: 192.168.3.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1
- DNS Server: 8.8.8.8

2.6 Test Connectivity



The image shows two screenshots of a Cisco Packet Tracer PC Command Prompt window. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the Command Prompt is open. The first screenshot shows a successful ping to 192.168.2.2 and 192.168.2.3. The second screenshot shows a failed ping to 192.168.3.2 and 192.168.3.3.

```
PC3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=6ms TTL=128
Reply from 192.168.2.2: bytes=32 time=2ms TTL=128
Reply from 192.168.2.2: bytes=32 time=2ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC3
Physical Config Desktop Programming Attributes
Command Prompt
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

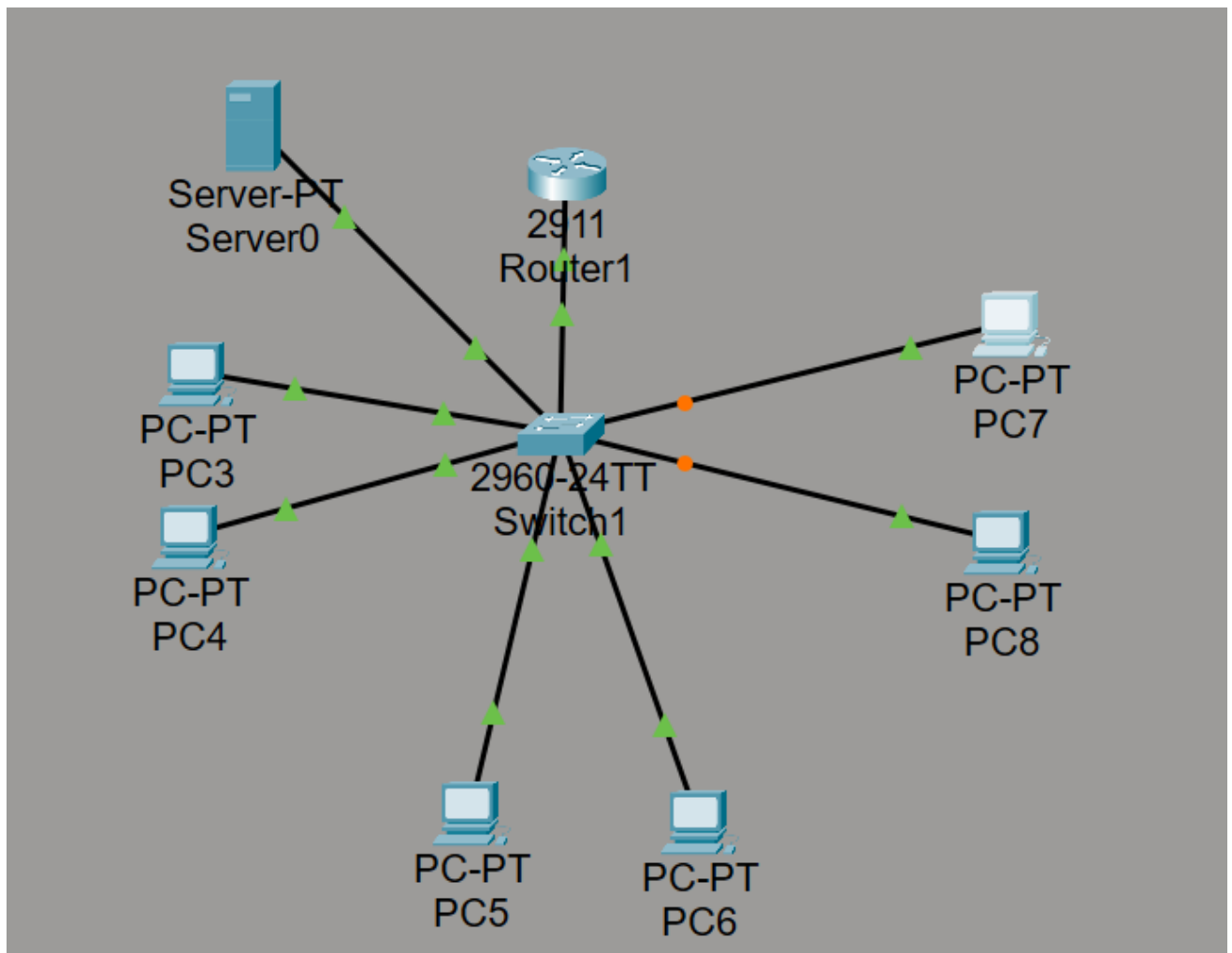
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time<1ms TTL=127
Reply from 192.168.3.3: bytes=32 time=3ms TTL=127
Reply from 192.168.3.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Part 3: Additional Task - Add new VLAN



3.1 Create VLAN5

```
Switch#
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 5
Switch(config-vlan)#name VLAN5
Switch(config-vlan)#exit
Switch(config)#
```

3.2 Assigning ports Fa0/7 and Fa0/8 to VLAN 5:

```
Switch(config)#int range f
Switch(config)#int range fastEthernet 0/7-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 5
Switch(config-if-range)#exit
Switch(config)#int fa
Switch(config)#int fastEthernet 0/1
Switch(config-if)#switchport trunk allowed vlan 2,3,4,5
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

3.3 Creating and configuring sub-interface and DHCP Relay:

```
Router(config)#int gigabitEthernet 0/0.5
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.5, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.5, changed state to up

Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#no shutdow
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

3.4. DHCP Configuration in the Server. Set up a DHCP pool for VLAN 5.

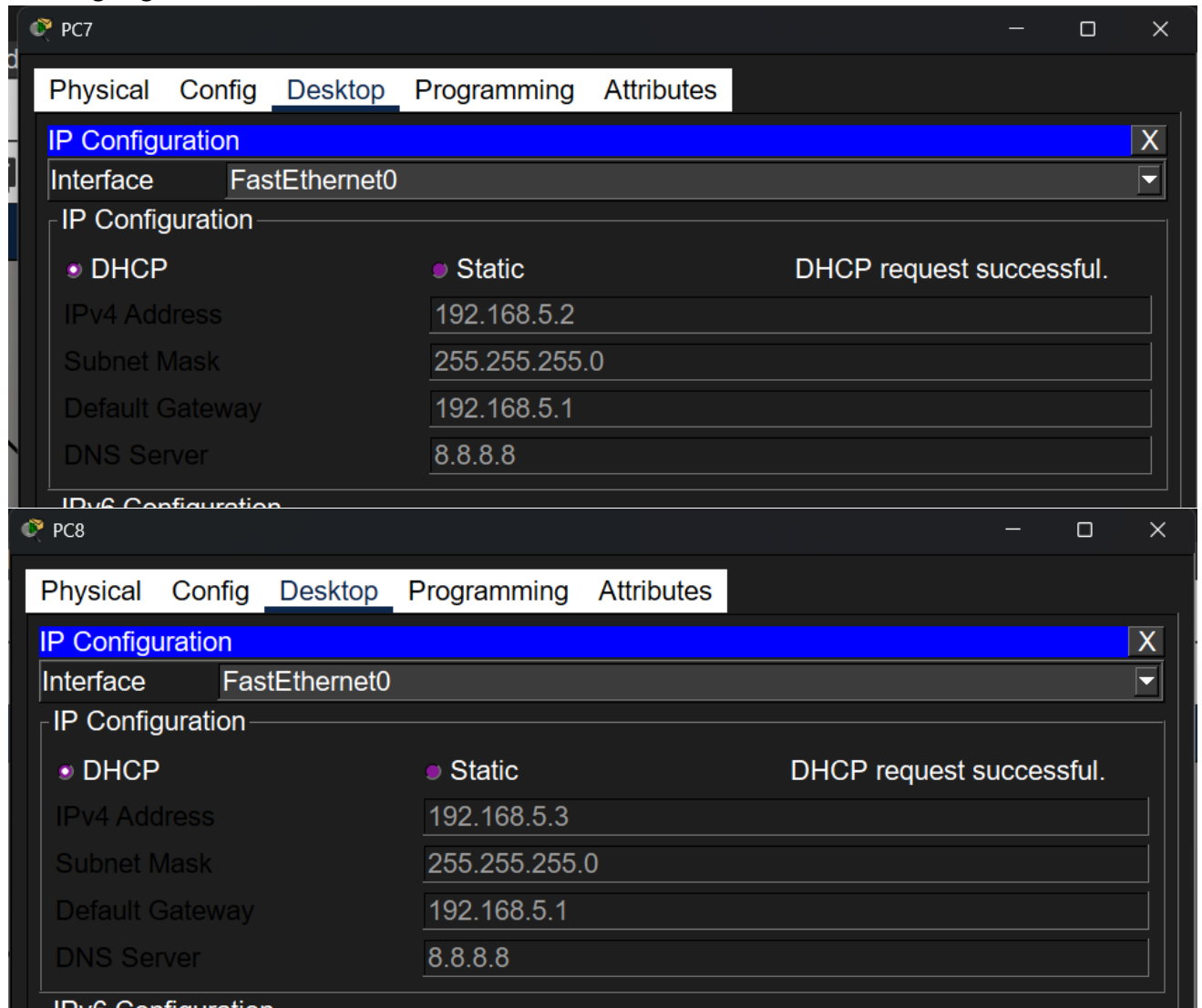
The screenshot shows the Cisco Packet Tracer interface for a server named 'Server0'. The 'Services' tab is selected, displaying the configuration for 'FastEthernet0'. The configuration includes a DHCP pool named 'DHCP-VLAN5' with the following settings:

- IP Address: 192.168.5.1
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.5.1
- DNS Server: 8.8.8.8
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

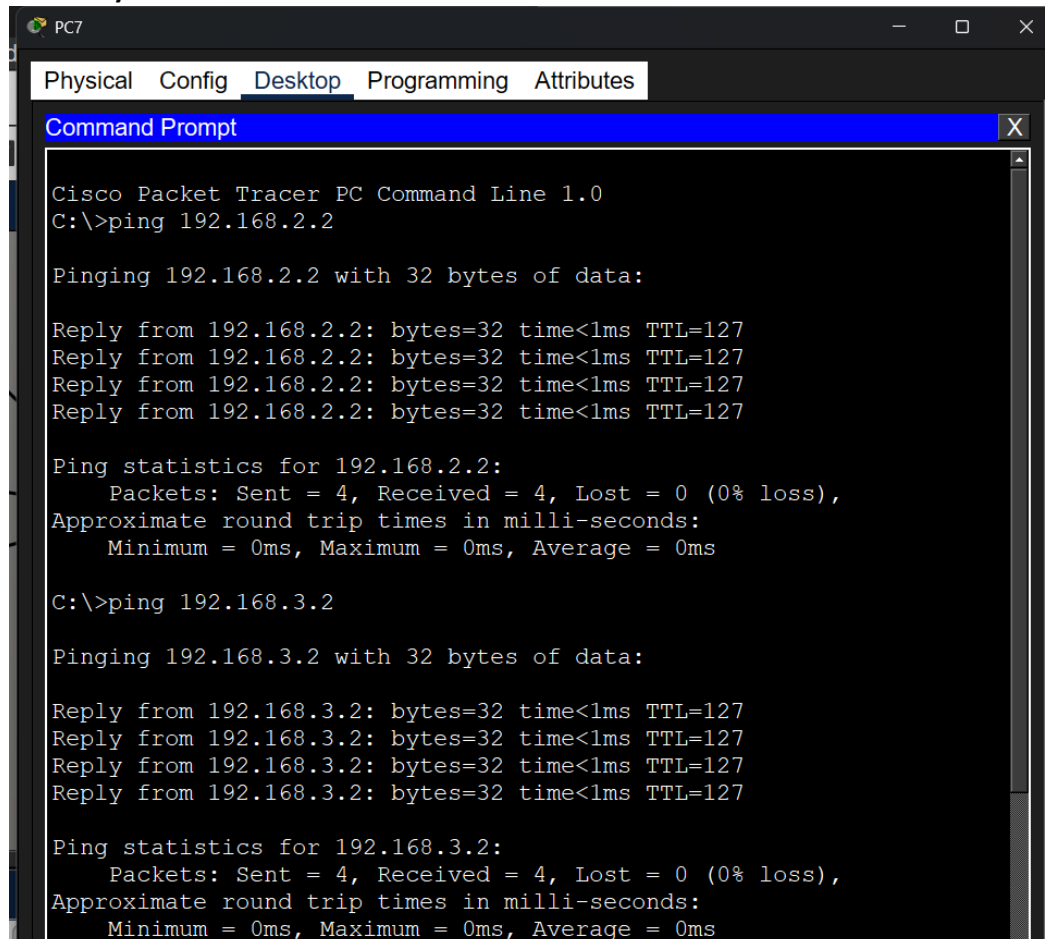
Below the configuration fields, there is a table showing the DHCP pool configuration:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
DHCP-...	192....	8.8.8.8	192....	255....	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192....	255....	512	0.0.0.0	0.0.0.0

3.5 Assigning IP to PC via DHCP



3.6 Verify that devices in VLAN 5 receive IP addresses via DHCP and can communicate with other VLANs.



```
C:\>ping 192.168.5.3
```

```
Pinging 192.168.5.3 with 32 bytes of data:
```

```
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.5.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>|
```